

## Questions with Answer Keys

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## Q1 - Single Correct

Suppose  $\left| \frac{f'(x)}{f''(x)} \cdot \frac{f(x)}{f'(x)} \right| = 0$ , where  $f(x)$  is continuously differentiable function with  $f'(x) \neq 0$  and satisfies  $f(0) = 1$

and  $f'(0) = 2$ , then  $f(x)$  is equal to

(1)  $x^2 + x + 1$

(2)  $2e^x - 1$

(3)  $e^{2x}$

(4)  $4e^{x/2} - 3$

## Q2 - Single Correct

If  $\int \frac{\cos x + \sin x + 1 - x}{e^x + \sin x + x} dx = \log_e[f(x)] + g(x) + C$ , where  $C$  is constant of integration and  $f(x)$  is positive, then  $f(x) + g(x)$

has the value

(1)  $e^x + \sin x + 2x$

(2)  $e^x + \sin x$

(3)  $e^x - \sin x$

(4)  $e^x + \sin x + x$

## Q3 - Single Correct

$\int x \cdot \frac{\log(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$  is equal to

(1)  $\sqrt{1+x^2} \cdot \log_e(x + \sqrt{1+x^2}) + C$

(2)  $\sqrt{x^2+1} \cdot \log(x + \sqrt{1+x^2}) + x + C$

(3)  $\sqrt{x^2+1} \log_e(x + \sqrt{1+x^2}) - x + C$

(4)  $\frac{x}{2} \log^2(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$

## Q4 - Single Correct

$\int \frac{(2x+1)}{(x^2+4x+1)^{3/2}} dx$  is equal to

(1)  $\frac{x}{(x^2+4x+1)^{1/2}} + C$

(2)  $\frac{x^3}{(x^2+4x+1)^{1/2}} + C$

(3)  $\frac{x^2}{(x^2+4x+1)^{1/2}} + C$

(4)  $\frac{1}{(x^2+4x+1)^{1/2}} + C$

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Q5 - Single Correct

$\int \sin(101x) \cdot \sin^{99} x dx$  is equal to

- (1)  $\frac{\sin(100x) \cdot (\sin x)^{100}}{100} + C$
- (2)  $\frac{\cos(100x) \cdot (\sin x)^{100}}{100} + C$
- (3)  $\frac{\cos(100x) \cdot (\cos x)^{100}}{100} + C$
- (4)  $\frac{\sin(100x) \cdot (\sin x)^{101}}{101} + C$

Q6 - Single Correct

Let  $f$  be a continuous function satisfying  $f'(\log_e x) = \begin{cases} 1, & \text{for } 0 < x \leq 1 \\ x, & \text{for } x > 1 \end{cases}$  and  $f(0) = 0$ , then  $f(x)$  can be defined as

- (1)  $f(x) = \begin{cases} 1, & \text{if } x \leq 0 \\ 1 - e^x, & \text{if } x > 0 \end{cases}$
- (2)  $f(x) = \begin{cases} 1, & \text{if } x \leq 0 \\ e^x - 1, & \text{if } x > 0 \end{cases}$
- (3)  $f(x) = \begin{cases} x, & \text{if } x < 0 \\ e^x, & \text{if } x > 0 \end{cases}$
- (4)  $f(x) = \begin{cases} x, & \text{if } x \leq 0 \\ e^x - 1, & \text{if } x > 0 \end{cases}$

Q7 - Single Correct

$\int \frac{px^{p+2q-1} - qx^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$  is equal to

- (1)  $-\frac{x^p}{x^{p+q}+1} + C$
- (2)  $\frac{x^q}{x^{p+q}-1} + C$
- (3)  $-\frac{x^q}{x^{p+q}+1} + C$
- (4)  $\frac{x^p}{x^{p+q}+1} + C$

Q8 - Single Correct

If  $I_n = \int (\sin x)^n dx$ ,  $n \in N$ , then  $(5/4 - 6I_6)$  is equal to

- (1)  $\sin x \cdot (\cos x)^5 + C$
- (2)  $\sin 2x \cdot \cos 2x + C$
- (3)  $\frac{\sin 2x}{8} \cdot (\cos^2 2x + 1 - 2 \cos 2x) + C$
- (4) None of the above

Q9 - Single Correct

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$\int \frac{e^{\tan^{-1}x}}{1+x^2} \left[ \left( \sec^{-1} \sqrt{1+x^2} \right)^2 + \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) \right] dx$ , when  $x > 0$ , is equal to

- (1)  $e^{\tan^{-1}x} \cdot \tan^{-1}x + C$
- (2)  $\frac{e^{\tan^{-1}x} \cdot (\tan^{-1}x)^2}{2} + C$
- (3)  $e^{\tan^{-1}x} \cdot \left[ \sec^{-1} \left( \sqrt{1+x^2} \right) \right]^2 + C$
- (4)  $e^{\tan^{-1}x} \cdot \left[ \operatorname{cosec}^{-1} \left( \sqrt{1+x^2} \right) \right]^2 + C$

Q10 - Single Correct

$\int \prod_{r=1}^n (x+r) \left( \sum_{k=1}^n \frac{1}{x+k} \right) dx$  is equal to

- (1)  $\prod_{r=1}^n (x+r)$
- (2)  $\sum_{k=1}^n \frac{1}{x+k}$
- (3)  $\sum_{r=1}^n \log(x+r)$
- (4) None of these

Q11 - Single Correct

If  $I_n = \int \cot^n x dx$ , then  $I_0 + I_1 + 2(I_2 + I_3 + \dots + I_8) + I_9 + I_{10}$  equals where,  $u = \cot x$

- (1)  $u + \frac{u^2}{2} + \dots + \frac{u^9}{9}$
- (2)  $-\left(u + \frac{u^2}{2} + \dots + \frac{u^9}{9}\right)$
- (3)  $-\left(u + \frac{u^2}{2!} + \dots + \frac{u^9}{9!}\right)$
- (4)  $\frac{u}{2} + \frac{2u^2}{3} + \dots + \frac{9u^9}{10}$

Q12 - Single Correct

$\int \frac{x^2(1-\log_e x)dx}{\log^4 x - x^4}$  is equal to

- (1)  $\frac{1}{2} \log \left( \frac{x}{\log x} \right) - \frac{1}{4} \log (\log^2 x - x^2) + C$
- (2)  $\frac{1}{4} \log \left( \frac{\log x + x}{\log x - x} \right) + \frac{1}{2} \tan^{-1} \left( \frac{\log x}{x} \right) + C$
- (3)  $\frac{1}{4} \log \left( \frac{\log x - x}{\log x + x} \right) - \frac{1}{2} \tan^{-1} \left( \frac{\log x}{x} \right) + C$
- (4) None of the above

Q13 - Single Correct

$\int \left( \frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta} \right) d\theta$  is equal to

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(1)  $\frac{1}{2} \log \left| \frac{\sec 27\theta}{\sec \theta} \right| + C$

(2)  $\frac{1}{2} \log \left| \frac{\sec \theta}{\sec 27\theta} \right| + C$

(3)  $\frac{1}{2} \log \left| \frac{2\sqrt{\sec 27\theta}}{\sec \theta} \right| + C$

(4) None of these

Q14 - Single Correct

If  $I = \int \frac{(x^2+20)}{(x \sin x + 5 \cos x)^2} dx$  is equal to

(1)  $-\frac{x}{\cos x(x \sin x + 5 \cos x)} + \tan x + C$

(2)  $\frac{x}{\sin x(x \sin x + 5 \cos x)} + C$

(3)  $(x \sin x + 5 \cos x)^{-1} \cdot \sin x + 7x + C$

(4) None of the above

Q15 - Single Correct

If  $I = \int \frac{x^n}{1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}} dx$ , then

(1)  $I = \frac{1}{n!} \left\{ x + \log \left( 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) \right\} + C$

(2)  $I = \frac{x^2 - e^x \log x}{n!(e^x - 1)} + C$

(3)  $I = n! \log \left( \frac{e^x}{1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}} \right) + C$

(4) None of the above

Q16 - Multiple Correct

$\int e^{3x} \cdot \cos 4x dx = e^{3x} (A \sin 4x + B \cos 4x) + C$ , then

(1)  $4A = 3B$

(2)  $2A = 3B$

(3)  $3A = 4B$

(4)  $4A + 3B = 1$

Q17 - Multiple Correct

Let  $f'(x) = 3x^2 \cdot \sin\left(\frac{1}{x}\right) + x \cos\left(\frac{1}{x}\right)$ ,  $x \neq 0$ ,  $f(0) = 0$ ,  $f\left(\frac{1}{\pi}\right) = 0$ , then which of the following is/are not correct?

(1)  $f(x)$  is continuous at  $x = 0$

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(2)  $f(x)$  is not differentiable at  $x = 0$

(3)  $f'(x)$  is discontinuous at  $x = 0$

(4)  $f'(x)$  is differentiable at  $x = 0$

Q18 - Multiple Correct

$\int x e^x \sin x dx$  is equal to

(1)  $\frac{e^x}{2} \{x(\sin x - \cos x) + \cos x\} + C$

(2)  $\operatorname{Im} \int x e^{(1+i)x} dx$

(3)  $\operatorname{Im} \left\{ \frac{x e^{(1+i)x}}{1+i} - \frac{e^{(1+i)x}}{(1+i)^2} \right\} + C$

(4)  $\operatorname{Re} \int x e^{(1+i)x} dx$

Q19 - Multiple Correct

If  $u = \int e^{ax} \cdot \cos bx dx$  and  $v = \int e^{ax} \cdot \sin bx dx$ , then

(1)  $(a^2 + b^2)(u^2 + v^2) = e^{2ax}$

(2)  $(a^2 - b^2)(u^2 - v^2) = e^{2ax}$

(3)  $\tan^{-1}\left(\frac{v}{u}\right) + \tan^{-1}\left(\frac{b}{a}\right) = bx$

(4)  $\tan^{-1}\left(\frac{v}{u}\right) - \tan^{-1}\left(\frac{b}{a}\right) = ax$

Q20 - Multiple Correct

Primitive of  $\sqrt{1 + 2 \tan x (\sec x + \tan x)}$  w.r.t.  $x$  is

(1)  $\log |\sec x| - \log |\sec x + \tan x| + C$

(2)  $\log |\sec x + \tan x| + \log |\sec x| + C$

(3)  $2 \log \left| \sec \frac{x}{2} + \tan \frac{x}{2} \right| + C$

(4)  $\log |1 + \tan x (\sec x + \tan x)| + C$

Q21 - Multiple Correct

$\int \frac{1}{x^2 - 1} \log \left| \frac{x-1}{x+1} \right| dx$  is equal to

(1)  $\frac{1}{2} \log^2 \left| \frac{x-1}{x+1} \right| + C$

(2)  $\frac{1}{4} \log^2 \left| \frac{x-1}{x+1} \right| + C$

(3)  $\frac{1}{2} \log^2 \left| \frac{x+1}{x-1} \right| + C$

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(4)  $\frac{1}{4} \log^2 \left| \frac{x+1}{x-1} \right| + C$

Q22 - Multiple Correct

$\int \frac{dx}{\sqrt{x-x^2}}$ , where  $x \in \left(\frac{1}{2}, 1\right)$  is equal to

(1)  $2 \sin^{-1} \sqrt{x} + C$

(2)  $\sin^{-1}(2x-1) + C$

(3)  $-\cos^{-1}(2x-1) + C$

(4)  $\cos^{-1}\left(2\sqrt{x-x^2}\right) + C$

Q23 - Multiple Correct

$\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$  is equal to

(1)  $\cot^{-1}(\cot^2 x) + C$

(2)  $-\cot^{-1}(\tan^2 x) + C$

(3)  $\tan^{-1}(\tan^2 x) + C$

(4)  $-\tan^{-1}(\cos 2x) + C$

Q24 - Multiple Correct

If  $I = \int \frac{dx}{x^r(1+x^r)^{1/r}}$ ,  $r \in N$  and  $I = \alpha(1+x^{-3})^\beta + C$ , then the quadratic equation having roots  $\alpha$  and  $\beta$  is

(1)  $6x^2 - x - 2 = 0$

(2)  $6x^2 - x + 2 = 0$

(3)  $6x^2 + x - 2 = 0$

(4)  $6x^2 + x + 2 = 0$

Q25 - Multiple Correct

$\int \frac{x^6(x+1)}{\sqrt{5x^{10}+6x^9+x^4}} dx$  is equal to

(1)  $\frac{\sqrt{5x^6+6x^5+1}}{15} + C$

(2)  $\frac{\sqrt{5x^8+6x^7+x^2}}{30} + C$

(3)  $\frac{\sqrt{5x^8+6x^7+x}}{15} + C$

(4)  $(5x^{10} + 6x^5 + x^4)^{1/2} + C$

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## Q26 - Multiple Correct

The value of  $\int \sqrt{x-3} \{ \sin^{-1}(\log_e x) + \cos^{-1}(\log_e x) \} dx$  is

(1)  $\frac{\pi}{3} (x-3)^{3/2}$

(2)  $\frac{\pi}{3} (x)^{3/2}$

(3)  $\frac{\pi}{2} (x-3)^{3/2}$

(4) Does not exist

## Q27 - Multiple Correct

$\int \frac{\operatorname{cosec}^2 x - 2011}{(\cos x)^{2011}} dx$  is equal to

(1)  $-\frac{\cot x}{(\cos x)^{2011}} + C$

(2)  $\frac{\cot x}{(\cos x)^{2010}} + C$

(3)  $-\frac{\tan x}{(\cos x)^{2011}} + C$

(4)  $\frac{\cot x}{(\sin x)^{2011}} + C$

## Q28 - Multiple Correct

Let  $P = \int e^{ax} \cdot \cos bxdx$  and  $Q = \int e^{ax} \cdot \sin bxdx$ . If  $P + iQ = Z + C$ , where  $Z$  is a complex function and  $C$  is constant of integration, then  $|Z|$  is equal to

(1)  $e^{ax}$

(2)  $\frac{e^{ax}}{a+b}$

(3)  $\frac{e^{ax}}{\sqrt{a^2+b^2}}$

(4)  $\frac{e^{bx}}{\sqrt{a^2+b^2}}$

## Q29 - Paragraph 1

## Passage I (For Question 29, 30)

Reduction formula make it possible to reduce an integral dependent on the index,  $n > 0$  called the order of the integral, to an integral of the same type with a smaller index. Integration by parts helps us to derive reduction formulas.

If  $I_{n,-m} = \int \frac{\sin^n x}{\cos^m x} dx$ , then  $I_{n,-m} + \frac{n-1}{m-1} I_{n-2,2-m}$  is equal to

(1)  $\frac{\sin^{n-1} x}{\cos^{m-1} x}$

(2)  $\frac{1}{m-1} \cdot \frac{\sin^{n-1} x}{\cos^{m-1} x}$



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(3)  $\frac{1}{n-1} \cdot \frac{\sin^{n-1} x}{\cos^{m-1} x}$

(4)  $\frac{n-1}{m-1} \cdot \frac{\sin^{n-1} x}{\cos^{m-1} x}$

Q30 - Paragraph 1

If  $u_n = \int \frac{x^n}{\sqrt{ax^2+2bx+c}} dx$ , then  $(n+1)au_{n+1} + (2n-1)bu_n + nc \cdot u_{n-1}$  equals

(1)  $x^{n-1}\sqrt{ax^2+2bx+c}$

(2)  $\frac{x^{n-2}}{\sqrt{ax^2+2bx+c}}$

(3)  $\frac{x^n}{\sqrt{ax^2+2bx+c}}$

(4)  $x^n\sqrt{ax^2+2bx+c}$

Q31 - Paragraph 2

Passage II (For Question 31, 32)

A curve is represented parametrically by the equations  $x = e^t \cos t$  and  $y = e^t \sin t$ , where  $t$  is a parameter.

The relation between the parameter  $t$  and the angle  $\alpha$  between the tangent to the given curve and the  $X$ -axis is given by

(1)  $\frac{\pi}{2} - \alpha$

(2)  $\frac{\pi}{4} + \alpha$

(3)  $\alpha - \frac{\pi}{4}$

(4)  $\frac{\pi}{4} - \alpha$

Q32 - Paragraph 2

The value of  $\frac{d^2y}{dx^2}$  at the point  $t = 0$  is

(1) 1

(2) 2

(3) -2

(4) 3

Q33 - Paragraph 3

Passage III (For Question 33, 34)

In calculating a number of integrals, we had to use the method of integration by parts several times in succession. The result



could be obtained more rapidly and in a more concise form by using the so called generalised formula for integration by parts.

$$\int u(x)v(x)dx = u(x)v_1(x) - u'(x)v_2(x) + u''(x)v_3(x) - \cdots + (-1)^{n-1}u^{(n-1)}(x)v_n(x) - (-1)^{n-1} \cdot \int u^{(n)}(x) \cdot v_n(x)dx$$

where,  $v_1(x) = \int v(x)dx$ ,  $v_2(x) = \int v_1(x)dx$ ,  $\dots$ ,  $v_n = \int v_{n-1}(x)dx$  of course, we assume that all derivatives and integrals appearing in this formula exist. The use of the generalised formula for integration by parts is especially useful when calculating  $\int P_n(x)Q(x)dx$ , where  $P_n(x)$  is polynomial of degree  $n$  and the factor of  $Q(x)$  is such that it can be integrated successively  $(n+1)$  times.

If  $\int (x^3 - 2x^2 + 3x - 1) \cos 2x dx = \frac{\sin 2x}{4}u(x) + \frac{\cos 2x}{8}v(x) + C$ , then

(1)  $u(x) = x^3 - 4x^2 + 3x$

(2)  $u(x) = 2x^3 - 4x^2 + 3x$

(3)  $v(x) = 3x^2 - 4x + 3$

(4)  $v(x) = 6x^2 - 8x$

**Q34 - Paragraph 3**

If  $\int e^{2x} \cdot x^4 dx = \frac{e^{2x}}{2}f(x) + C$ , then  $f(x)$  is

(1)  $\frac{1}{2} \left( x^4 - 2x^3 + 3x^2 - 3x + \frac{3}{2} \right)$

(2)  $(x^4 - x^3 + 2x^2 - 3x + 2)$

(3)  $\left( x^4 - 2x^3 + 3x^2 - 3x + \frac{3}{2} \right)$

(4)  $\left( x^4 - 2x^3 + 2x^2 - 3x + \frac{3}{2} \right)$

**Q35 - Paragraph 4**

**Passage IV (For Question 35, 36)**

Let  $\lim_{x \rightarrow \infty} \frac{\int (\sin x + \sin ax)^2 dx}{x} = L$

For  $a = 0$ , then the value of  $L$  is

(1) 0

(2)  $\frac{1}{2}$

(3) 1

(4) 2

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Q36 - Paragraph 4

For  $a = 1$ , then the value of  $L$  is

- (1) 0
- (2)  $\frac{1}{2}$
- (3) 1
- (4) 2

Q37 - Integer Type

A curve  $g(x) = \int x^{27} \cdot (1 + x + x^2)^6 \cdot (6x^2 + 5x + 4) dx$  passing through origin, then  $|7g(-1)|$  is equal to

Q38 - Integer Type

Let  $f(xy) = f(x) \cdot f(y)$ ,  $\forall x > 0, y > 0$  and  $f(x+1) = 1 + x \cdot \{1 + g(x)\}$ , where  $\lim_{x \rightarrow 0} g(x) = 1$ . If

$\int \frac{f(x)}{f'(x)} dx = \frac{x^2}{k} + C$ , then  $k$  is equal to

Q39 - Integer Type

If  $\int e^{x \sin x + \cos x} \cdot \left( \frac{x^4 \cos^3 x - x \sin x + \cos x}{x^2 \cos^2 x} \right) dx = e^{x \sin x + \cos x} \cdot \frac{1}{x} [f(x)] + C$ , then  $|f(0)|$  is equal to

Q40 - Integer Type

Let  $A = \int \frac{dx}{x^2 + 6x + 25}$  and  $B = \int \frac{dx}{x^2 - 6x - 27}$ . If  $12(A + B) = \lambda \cdot \tan^{-1}\left(\frac{x+3}{4}\right) + \mu \log\left|\frac{x-9}{x+3}\right| + C$ , then the value of  $(\lambda + \mu)$  is

Q41 - Integer Type

Let  $\int \frac{dx}{x^{2008} + x} = \frac{1}{p} \log\left(\frac{x^q}{1+x^r}\right) + C$ , where  $p, q, r \in N$  and need not to be distinct, then the value of  $\left(\frac{p+q}{r}\right)$  is

Q42 - Integer Type

If  $\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx = \mu e^x \left(\frac{1+x}{1-x}\right)^\lambda + C$ , then  $2\lambda + \mu$  is equal to

Q43 - Integer Type

If  $\int \frac{dx}{x^{2010} \cdot (1+x^{2010})^{1/2010}} = -\frac{1}{a} (1+x^{-b})^{\frac{c-2}{b}} + k$ , then  $\left(\frac{a+b-c}{1004}\right)$  is equal to

Q44 - Integer Type

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If  $\int \frac{dx}{(1+\sqrt{x})^{2010}} = 2 \left[ \frac{1}{\alpha(1+\sqrt{x})^\alpha} - \frac{1}{\beta(1+\sqrt{x})^\beta} \right] + C$ , where  $\alpha, \beta > 0$ , then  $|\alpha - \beta|$  is equal to

## Q45 - Integer Type

If  $\int \frac{dx}{x^{1/2}(1+x^2)^{5/4}} = \frac{A\sqrt{x}}{\sqrt[4]{1+x^2}} + C$ , then  $A$  equals

## Q46 - Integer Type

If  $\int \frac{(2x+3)}{x(x+1)(x+2)(x+3)+1} dx = C - \frac{1}{f(x)}$ , where  $f(x)$  is of the form of  $ax^2 + bx + c$ , then  $(a + b + c)$  equals

## Q47 - Integer Type

If  $\int (\sqrt{\tan x} + \sqrt{\cot x}) dx = \sqrt{\lambda} \tan^{-1} \left( \frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{\mu}} \right) + C$ , then  $(\lambda + \mu)$  equals

Answer Key

|               |                  |               |               |
|---------------|------------------|---------------|---------------|
| Q1 (3)        | Q2 (2)           | Q3 (3)        | Q4 (1)        |
| Q5 (1)        | Q6 (4)           | Q7 (3)        | Q8 (3)        |
| Q9 (3)        | Q10 (1)          | Q11 (2)       | Q12 (3)       |
| Q13 (4)       | Q14 (1)          | Q15 (3)       | Q16 (3, 4)    |
| Q17 (2, 3, 4) | Q18 (1, 2, 3)    | Q19 (1, 3)    | Q20 (1, 2, 4) |
| Q21 (2, 4)    | Q22 (1, 2, 3, 4) | Q23 (1, 2, 3) | Q24 (1)       |
| Q25 (1)       | Q26 (4)          | Q27 (1)       | Q28 (3)       |
| Q29 (2)       | Q30 (4)          | Q31 (3)       | Q32 (2)       |
| Q33 (2)       | Q34 (3)          | Q35 (2)       | Q36 (4)       |
| Q37 (1)       | Q38 (4)          | Q39 (1)       | Q40 (4)       |
| Q41 (2)       | Q42 (2)          | Q43 (2)       | Q44 (1)       |
| Q45 (2)       | Q46 (5)          | Q47 (4)       |               |